

Question	Answer	Marks
6(a)	$2.4 / 6.0 = 8 / (R + 8)$	C1
	$R = 12 \Omega$	A1
	or $I = 2.4 / 8$ $= 0.30$ $R = (6 - 2.4) / 0.3$ or $R = (6 / 0.3) - 8$	(C1)
	$R = 12 \Omega$	(A1)
6(b)	$V = I(\rho L / A)$ or $V = IR$ and $R = (\rho L / A)$	M1
	I, ρ, A are constant (so $V \propto L$)	A1
6(c)(i)	$V_{XP} / V_{XY} = L_{XP} / L_{XY}$ $E / 2.4 = 1.24 / 2.00$	C1
	$E = 1.5 \text{ V}$	A1
6(c)(ii)	p.d. across XY / wire increases / p.d. across XP increases	M1
	so P moved towards X / away from Y / to the left	A1

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Question	Answer	Marks
3(c)	(i) $\frac{1}{2}$ (ii) $\frac{1}{2}$	20